

(1)
CLASS TEST
CO-4-AT-1

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COMPILER DESIGN

1. Construct a three address code for the following
if $(C(a < b) \text{ and } (C(c < d) \text{ or } (a > d)))$ then $Z = x + y * z$
else $Z = Z + 1$

(A) Step 1:- Identify the operation
1. $a < b$ 2. $c < d$ 3. $a > d$ 4. $(c < d) \text{ or } (a > d)$ 5. $(a < b)$
 $Z = x + y * z$
 $y * z \Rightarrow$ Since multiplication has higher priority.

Step 2:- Create temporaries for conditions

$$t_1 = a < b \quad t_2 = c < d \quad t_3 = a > d$$

$$(c < d) \text{ or } (a > d) \Rightarrow t_2 + t_3$$

$$t_4 = t_2 + t_3$$

$$(a < b) \text{ and } [(c < d) \text{ or } (a > d)]$$

$$t_5 = t_1 \text{ and } t_4$$

Step 3:- Generate TAC if-else Part

if t_5 goto L_1

goto L_2

Here:-

$\star L_1 \rightarrow$ Then $\star L_2 \rightarrow$ Else $L_3 \rightarrow$ End-of-if-else

Step 4: - Convert then Statement

$$Z = x + y * z$$

$$T_6 = y * z$$

$$T_7 = x + t_6$$

$$Z = t_7$$

goto L3

Step 5: - Generate TAC for else part
 $Z = Z + 1$

TAC:-

$$L_2: - t_8 = Z + 1$$

$$Z = t_8$$

then reach end:-

L3:

Final three address code:-

$$t_1 = a < b$$

$$t_2 = c < d$$

$$t_3 = a > d$$

$$t_4 = t_2 \text{ or } t_3$$

$$t_5 = t_1 \text{ and } t_4$$

if t_5 goto L₁

goto L₂

L₁:

$$t_8 = y * z$$

$$T_7 = x + t_6$$

$$Z = T_7$$

goto L₃

L₂:

$$t_8 = Z + 1$$

$$Z = t_8$$

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Three Address Code Using Backpatching
Given Boolean Expression

$(a < b) \text{ and } (c < d) \text{ or } (e < f)$

① STEP 1:- Apply Operator precedence
AND has higher precedence than OR, so
the expression is interpreted as
 $[(a < b) \text{ AND } (c < d)] \text{ OR } (e < f)$

STEP 2:- Generate incomplete jumps

1. if $a < b$ goto _
2. goto _
3. if $c < d$ goto _
4. goto _
5. if $e < f$ goto _
6. goto _

The underscore represents a target address that is known yet

STEP 3:- Handle AND

For $(a < b) \text{ AND } (c < d)$, if $a < b$ is true, control must continue to second condition

Backpatch statement 1 to statement 3

1. if $a < b$ goto 3

STEP 4:- Handle OR

If the first AND expression is false, the second OR operand $e < f$ must be evaluated

Steps:- Final backpatched code

1. if $a < b$ goto 3
2. goto 5
3. if $c < d$ goto 7
4. goto 5
5. if $e < f$ goto 7
6. goto 8
7. True
8. False

Construct the three address code using the following for production of boolean expressions

$B \rightarrow B_1 \mid B_2$ $B \rightarrow B_1 \text{ or } B_2$ $B \rightarrow B_1$ $B \rightarrow E_1 \text{ rel op } E_2$
 $B \rightarrow \text{true}$ $B \rightarrow \text{false}$

A) if $a > b$ goto _
goto _

if $c < d$ goto _
goto _

if $a > b$ goto true
goto 3

if $c < d$ goto true
goto 5

if $e < f$ goto true
goto False

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if $a > b$ goto 7

goto 3

if $c < d$ goto 7

goto 5

if $e < f$ goto 7

goto 8

True

False

4. Generate the tree Address Code for the following program fragment :-

While $(A \leq C \text{ and } B > d)$ do

if $A = 1$ then $C = C + 1$

else while $A \leq B$ do

$A = A + B$

(A) L_1 : if $A \leq C$ goto L_2
goto LEnd

L_2 : if $B > d$ goto L_3
goto LEnd

L_3 : if $A = 1$ goto L_4
goto L_5

L_4 : $t_1 = C$

$C = t_1$

goto L_1

L5: if $AC=0$ goto L6
goto L1

L6: $t_2 = A + B$
 $A = t_2$
goto L5
end

